



African Development Bank — Secretariat of STG17
 African Union — STATAFRIC
 SHaSA II · Specialized Technical Group 17 · Action Plan 2025–2030



Emerging Issues, Emerging Practice

Innovating the Data Value Chain — AI, LLMs and Big Data for Official Statistics

DATES

28 September – 2 October 2026

VENUE

Kigali, Rwanda

FORMAT

5 days · 27 sessions · 12 laboratories

Five days, from the concepts to a published repository. Every half-day pairs a short presentation with an extended laboratory at the keyboard.

Day 1 · AI concepts and the infrastructure behind them

Monday — Shared vocabulary, country experience, and what it really takes to build and run AI

Time	Session
09:00–09:15	Opening ceremony Welcome and opening remarks by the African Development Bank, AU STATAFRIC, and NISR. Objectives of the week, presentation of the STG17 Action Plan 2025–2030 and of where this workshop sits within it. Tour de table. CEREMONY
09:15–09:35	The AfDB Data Innovation for Africa (DI4A): Mission, current portfolio and upcoming milestones Why the Bank set up DI4A, what the current portfolio funds, and the milestones ahead — including what national statistical offices can expect to draw on over the next two years. PRESENTATION
09:35–10:05	The UN Big Data Regional Hub in Rwanda: milestones and use cases Ten years of the UN Committee of Experts on Big Data and Data Science for Official Statistics, and what the Regional Hub hosted by NISR has delivered since its launch: the use cases taken furthest PRESENTATION
10:05–10:30	The AI Family Tree: How the concepts fit together Building a shared concept map — AI, LLM, prompt engineering, RAG, fine-tuning, agentic systems, agents, MCP. What each concept can and cannot do for official statistics, and the vocabulary discipline that avoids costly misunderstandings. This half-hour sets the language used for the rest of the week. PRESENTATION
10:30–10:45	Coffee break
10:45–11:45	Country experiences in AI and the use of non-traditional big data Short country presentations on what has actually been attempted at home — pilots launched, data partnerships signed, obstacles met, results published. Countries are invited to be candid about what did not work; that is the more useful half of the exchange. PLENARY
11:45–12:30	Synthesis: where the continent stands Facilitated discussion structured around four questions drawn from the presentations — which use cases recur, which data partnerships are replicable, which obstacles are shared, and where pooling effort would pay. FACILITATED DISCUSSION
12:30–14:00	Lunch

Time	Session
14:00–14:30	AI Infrastructure: What it really takes to run AI in a Statistical Office Fundamentals and needs — GPUs and accelerators, memory and context, inference versus training, latency and throughput. Cloud, hybrid or on-premise; data sovereignty and confidentiality constraints for NSO microdata; cost modelling per use case; open-weight versus proprietary models. PRESENTATION
14:30–15:30	Hands-on part 1 — Retrieval-Augmented Generation Build a working RAG assistant over a statistical corpus — chunking, embeddings, vector store, retrieval, grounded answering. Test it against questions with known answers and observe where retrieval fails, and why. LABORATORY
15:30–16:45	Hands-on part 2 — From RAG to Agent Turn the retriever built before the break into a tool, and wrap it in an agent — tools and function calling, planning and iteration, memory, error handling and human-in-the-loop checkpoints. The agent answers a question, retrieves the supporting figures and drafts a short note; participants then examine exactly where it must stay supervised. LABORATORY
16:45–17:00	Coffee break

Day 2 · Prompt Engineering and the Optimised Use of LLMs

Tuesday — Getting reliable, fast and affordable work out of language models

Time	Session
09:00–10:30	Talking to Machines: The Craft of Prompt Engineering Task framing, role and context, explicit constraints, few-shot examples, decomposition, structured JSON outputs, guardrails and systematic iteration. Why models hallucinate and how prompt design reduces it. Common anti-patterns and how to detect them in your own prompts. PRESENTATION AND LABORATORY
10:30–10:45	Coffee break
10:45–12:30	Hands-on — From Statistical Document to Public Dashboard Take a national statistical publication (PDF or report), use an LLM to extract and structure its data, verify the extraction against the source, generate an interactive dashboard and publish it live on GitHub Pages. Each participant leaves with a public URL. LABORATORY
12:30–14:00	Lunch
14:00–14:45	From Good to Great: Prompt Optimisation Systematic improvement rather than trial and error — building an evaluation set, measuring output quality, controlling variance, managing context length and token cost, caching, and choosing between prompting, RAG and fine-tuning. PRESENTATION AND LABORATORY
14:45–15:30	Choosing your engine: speed, cost and sovereignty — working with Groq How inference speed changes what is feasible in a production statistical pipeline. Participants run the same prompt against two providers, including Groq, and record latency, cost per thousand documents and output quality in a shared comparison sheet. The sheet becomes a reusable procurement input for their office. BENCHMARK
15:30–16:45	Hands-on — One Model, Many Jobs: an LLM Toolkit for Statisticians Rotating stations across professional use cases — writing and debugging code; drafting reports and methodological notes; producing presentations; generating charts and graphics; image and visual identity work, including logo design; document analysis and audio briefings with NotebookLM; multimodal work with Gemini. Each participant selects two stations relevant to their office and leaves with a deliverable from each. LABORATORY
16:45–17:00	Coffee break

Day 3 · Beyond the Survey: Non-Traditional Data and the Technologies That Handle Them

Wednesday — New raw material for official statistics — and the engines that make it usable at scale

Time	Session
09:00–10:30	Non-Traditional Data Sources: The Hidden Treasure Taxonomy of non-traditional sources — satellite imagery, crowdsourced measurement, scanner and transaction data, web scraping, sensors and IoT, citizen-generated data. Quality frameworks and coverage bias in non-probabilistic sources; access and partnership models with private data holders; ethics and confidentiality. PRESENTATION AND LABORATORY
10:30–10:45	Coffee break
10:45–12:30	Hands-on — Ookla Speedtest Open Data and WorldPop Ookla: performance tiles at web-Mercator zoom 16 (about 611 m at the equator), quarterly since Q1 2019; key fields (quadkey, avg_d_kbps, avg_lat_ms, tests, devices); CC BY-NC-SA 4.0 licence and its consequences for NSO publication; verify national coverage before designing any indicator. WorldPop: gridded population rasters used to weight and normalise. Teams join the tiles to administrative boundaries and produce a first population-weighted connectivity indicator per region. LABORATORY
12:30–14:00	Lunch
14:00–14:30	Engines of Scale: Big Data Technologies for Statistical Systems Fundamentals — ingestion, storage (data lake and lakehouse), distributed processing (batch versus streaming, Spark), indexing and search, orchestration, metadata and versioning. PRESENTATION
14:45–15:30	Hands-on part 1 — Ookla at Scale with Elasticsearch Index the Ookla tiles in a pre-provisioned Elasticsearch cluster; understand the mapping and the geo_shape type; run the first aggregation and geospatial queries. LABORATORY
15:30–16:45	Hands-on part 2 — Search-driven exploration Build a search-driven exploration of connectivity by region and quarter; compare query time and memory footprint against the pandas approach of the morning, and draw the practical conclusion about when the added complexity is worth it. Publication on GitHub Each team commits its notebooks, queries and outputs to the workshop GitHub organisation, with a README documenting method, sources and licence. LABORATORY
16:45–17:00	Coffee break

Day 4 · Africa after dark: Reading development from Night-Time Lights

Thursday — A full day on NTL — from raw radiance to a validated subnational indicator

Time	Session
09:00–09:30	Night-Time Lights: What the darkness tells us Concepts and utility, kept deliberately short so the day is spent in the data — from DMSP-OLS (1992–2013) to VIIRS/DNB; the NASA Black Marble products (VNP46A2 daily, A3 monthly, A4 annual); what NTL proxies well — economic activity, electrification, urbanisation, crisis monitoring. The known artefacts are introduced here by name only and met hands-on in Part 2. PRESENTATION
09:30–10:30	Hands-on part 1 — Collect Access NTL rasters (NASA Earthdata, EOG or Google Earth Engine); understand the file structure, bands and quality flags; clip to the national extent and save a working subset. LABORATORY
10:30–10:45	Coffee break

Time	Session
10:45–12:30	Hands-on part 2 — Explore and Understand Visualise and interrogate the raster before computing on it — distribution of radiance values, effect of the cloud and quality masks, month-to-month variation. Teams deliberately hunt for the artefacts named in the morning talk (blooming around cities, saturation, gas flares, seasonality, sensor discontinuity, rural low-light noise) and document which ones are present in their own country. Closes with a short round of comparisons across teams. LABORATORY
12:30–14:00	Lunch
14:00–16:45	Hands-on part 3 — NTL Analysis & Validation Compute zonal statistics by administrative level (sum of radiance, mean radiance, lit area); build annual and monthly time series; detect change between two periods; map and visualise the results. Then validate the NTL proxy by correlating it with official subnational statistics (population, electrification rate) and decide, with evidence, whether it is usable for dissemination in your country or remains a diagnostic tool only. Document the limitations explicitly; this statement is part of the deliverable. The resulting maps, charts and limitations statement can be published as a public website via GitHub Pages. LABORATORY
16:45–17:00	Coffee break

Day 5 · Shared Resources, Country Results and the Way Forward

Friday (morning only) — Turning a week of practice into a continental resource and into commitments under the Action Plan

Time	Session
09:00–10:30	Assisting countries in publishing their work How a national team takes what it built this week to a durable public product — repository structure and README conventions, GitHub Pages, dataset and code licensing, DOI and citation, metadata, versioning and maintenance responsibility. LABORATORY
10:30–10:45	Coffee break
10:45–12:15	Country presentations Each team presents the work produced during the week — dashboard, connectivity scores, NTL analysis — with its GitHub repository and an honest statement of limitations. Teams are formed at country or sub-regional level so that the slot holds. PLENARY
12:15–12:30	Key takeaways and way forward Consolidation of the wall board kept open all week, and agreement on the follow-up webinar topics. PLENARY
12:30–12:45	Closing ceremony Closing remarks by the African Development Bank and AU STATAFRIC. CEREMONY

Full detail, materials and notebooks: <https://datainnov4africa-web.github.io/stg17-workshop/>